

BEOGRAD BATAJNICA, Serbia

WMO: 132725

Lat: 44.935N

Lon: 20.258E

Elev: 81

StdP: 100.36

Time Zone: 1.00 (EUC)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-10.9	-8.1	-13.8	1.1	-9.8	-10.8	1.5	-7.2	10.4	1.0	9.4	2.2	2.9	320	0.474

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	13.8	35.1	21.4	33.2	21.3	31.8	21.0	23.4	30.5	22.5	30.1	21.6	29.2	3.3	100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.2	16.0	26.2	20.1	15.0	25.2	19.2	14.1	24.0	70.0	30.5	66.4	30.3	63.6	29.6	26.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.4	8.1	7.0	DB	-16.7	37.0	5.1	2.1	-20.3	38.5	-23.3	39.8	-26.1	41.0	-29.8	42.5	(4)
(5)			WB	-16.9	25.0	5.0	1.0	-20.5	25.7	-23.5	26.3	-26.3	26.8	-30.0	27.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	12.5	0.5	2.9	7.6	13.1	17.1	21.1	22.9	23.0	18.3	12.7	8.2	2.3		
	DBStd	8.82	5.08	5.14	4.38	3.64	3.53	3.29	2.99	3.13	3.97	4.07	4.47	4.74		
	HDD10.0	955	296	201	98	12	1	0	0	0	1	21	84	242		
	HDD18.3	2589	553	432	333	160	66	12	3	3	48	178	303	498		
	CDD10.0	1877	1	2	23	106	221	333	400	404	250	104	31	2		
	CDD18.3	470	0	0	0	3	27	95	144	149	48	3	0	0		
	CDH23.3	5519	0	0	2	86	315	1054	1677	1778	542	65	1	0		
	CDH26.7	2328	0	0	0	16	65	419	768	863	191	6	0	0		
(14) Wind	WSAvg	2.9	3.0	3.4	3.4	3.3	2.8	2.5	2.4	2.3	2.5	2.7	3.1	2.9		
(15) Precipitation	PrecAvg	667	44	39	46	50	66	84	67	53	63	54	48	52		
	PrecMax	1011	85	89	118	148	244	175	235	128	187	115	134	140		
	PrecMin	352	8	1	6	3	17	18	11	3	10	0	3	5		
	PrecStd	159	21	21	32	29	43	46	55	35	36	33	28	31		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.9	19.1	23.1	28.9	30.8	34.2	37.0	37.9	34.0	27.1	22.1	16.1		
		MCWB	10.8	11.5	13.2	17.4	19.7	21.8	21.9	21.2	20.4	17.4	15.4	11.9		
	2%	DB	11.8	15.2	20.9	26.0	28.2	32.9	35.0	35.9	31.2	25.2	19.9	13.2		
		MCWB	8.9	10.1	11.8	16.0	18.5	22.3	21.7	21.0	19.2	16.5	14.1	10.1		
	5%	DB	9.2	12.8	18.2	23.2	26.9	31.0	33.0	33.8	28.8	23.0	17.1	11.0		
		MCWB	7.7	8.7	10.9	14.2	18.1	21.3	21.6	20.9	18.5	15.6	12.8	8.4		
	10%	DB	7.2	10.1	15.8	21.1	25.0	29.1	31.0	31.8	26.2	20.2	15.0	8.8		
		MCWB	6.1	7.5	9.6	13.5	17.3	20.5	21.1	20.7	17.6	14.8	11.7	7.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.9	12.1	14.2	18.1	20.6	24.9	24.3	24.1	21.4	18.7	15.9	12.7		
		MCDB	13.8	18.1	21.9	26.9	28.4	31.6	31.8	29.9	31.4	24.9	21.2	14.6		
	2%	WB	9.4	10.5	12.5	16.4	19.5	23.2	23.4	22.8	20.3	17.3	14.4	10.5		
		MCDB	11.1	14.5	18.7	23.9	26.7	30.8	30.9	28.2	23.4	18.9	13.1			
	5%	WB	7.8	9.1	11.3	15.2	18.6	22.0	22.5	22.1	19.3	16.2	13.1	8.8		
		MCDB	9.1	12.0	17.0	21.5	25.3	28.3	30.2	30.5	26.4	21.7	16.5	10.9		
	10%	WB	6.1	7.7	10.2	14.1	17.7	21.1	21.6	21.3	18.4	15.1	11.9	7.1		
		MCDB	7.0	9.7	14.7	19.6	23.7	27.4	29.2	29.4	24.8	19.9	14.7	8.1		
(35) Mean Daily Temperature Range	5% DB	MDBR	6.9	8.5	10.9	12.0	11.5	11.8	13.0	13.8	12.2	11.4	9.3	6.9		
		MCDBR	10.7	13.8	16.4	16.6	14.5	15.1	16.4	17.6	16.2	15.8	12.7	11.1		
	5% WB	MCWBR	8.2	8.8	8.9	8.0	6.5	6.0	5.7	6.0	6.1	8.1	7.6	8.2		
		MCWBR	10.0	12.5	14.3	13.9	13.3	13.3	14.1	14.5	13.6	14.7	12.0	10.7		
(40) Clear-Sky Solar Irradiance	taub		0.348	0.374	0.406	0.463	0.450	0.445	0.449	0.457	0.437	0.401	0.363	0.345		
	taud		2.332	2.247	2.206	2.056	2.158	2.208	2.194	2.171	2.196	2.292	2.353	2.350		
	Ebn at Noon		745	791	818	798	824	828	819	796	776	752	720	706		
	Edh at Noon		81	105	125	157	147	140	141	139	125	99	79	72		
(44) All-Sky Solar Radiation	RadAvg		1.23	2.06	3.29	4.54	5.54	6.13	6.21	5.51	3.85	2.65	1.60	1.03		
	RadStd		0.17	0.28	0.40	0.46	0.52	0.56	0.47	0.53	0.42	0.29	0.14	0.15		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (51 neighbors)	+0.67	N/A	N/A	N/A	+0.36	N/A	-116	-179	+118	+61

Nomenclature: See separate page